

ECSE 2500

Jan. 26

Lec 5

Topic: Probability in continuous space and Conditional Probability

□ Probability in Continuous Sample Space

Because $P(\text{Exact one outcome}) = 0$ in Continuous Space

a natural idea is to assign based on the interval, e.g., X continuous number $[0, 1]$,

Define $P(a \leq x \leq b) = b - a$, for $0 \leq a \leq b \leq 1$.

This is called the uniform distribution. ^{Axiom 1}

It says that the probability of getting a value in an interval equals the width/length of that interval. Therefore,

$$P(\underbrace{[0, 1]}_S) = 1 - 0 = 1 \quad \checkmark \quad \text{Validates Axiom 2} \quad P([0, \frac{1}{2}]) = \frac{1}{2}$$

More example on uniform distribution, which can be evaluated geometrically.

Example

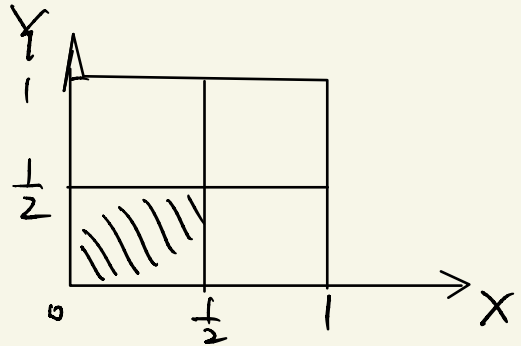
Pick two numbers X and Y .

Each of them is between 0 and 1.

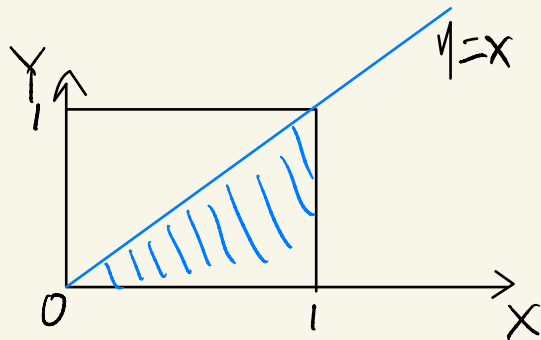
What is

$$P\left(X \leq \frac{1}{2}, Y \leq \frac{1}{2}\right)?$$

$$= \frac{1}{4}$$



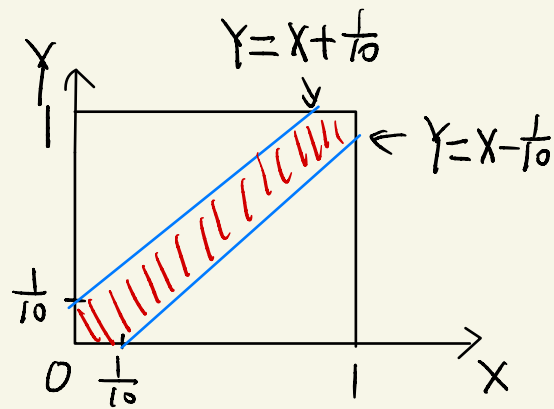
$$P(X \geq Y)$$
$$= \frac{1}{2}$$



$$P(|X - Y| < \frac{1}{10}) ?$$

$$= 1 - 2 \times \underbrace{\left(\frac{9}{10}\right)^2 \times \frac{1}{2}}_{\frac{81}{100}}$$

$$= \frac{19}{100}$$



○ An interesting example on discrete sample space

The birthday problem

$P(\text{at least 2 students of a class of 60 students have the same birthday}) ?$

$$= 1 - P(\text{no 2 students share the same birthday})$$

$$= 1 - \frac{\text{all combination of 60 different birthdays}}{\text{All possible combination of 60 birthdays}}$$

$$= 1 - \frac{365 \cdot 364 \cdot 363 \cdots (365 - 60 + 1)}{365^{60}}$$

$$= 1 - 0.0008 = 0.9992 \leftarrow \text{large probability}$$

□ Introduce Conditional Probability

Motivation: We often have encountered problems of the following form, "What is the probability of X , given that I observed that Y happens?"

Any kind of diagnosis problems where we see some symptoms and want to figure out the causes/disease will have the conditional probability

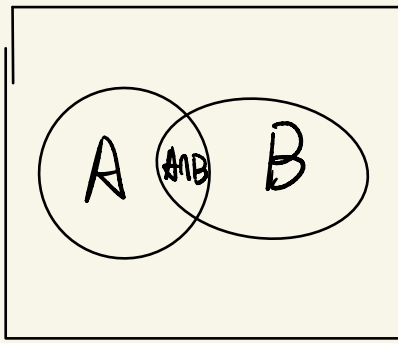
$$P(\text{Covid-19} \mid \text{Test kit is positive})$$

Mathematical definition

We have two events A and B , and we define the notation

$$P(A \mid B)$$

to measure the probability of event A given that the event B happens.



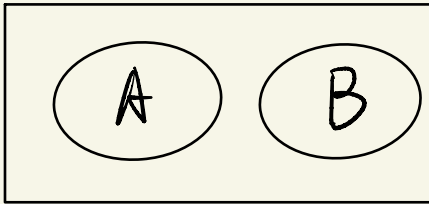
We ~~define~~ that

$$P(A|B) \stackrel{\text{def}}{=} \frac{P(A \cap B)}{P(B)}$$

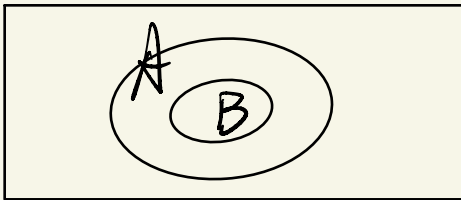
(as long as $P(B) > 0$)

This makes sense; It relates the sample space of event B to A and normalize so that $P(B) = 1$.

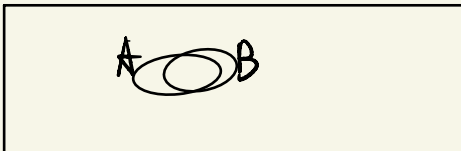
How likely is $A \cap B$ relative to B?



$$P(A|B) = 0$$



$$P(A|B) = 1$$



$P(A), P(B)$ both small,
but $P(A|B)$ is large

Rare disease

Rare symptom

Or, we may get very little information about Event A given Event B.

$$P(A|B) = P(A)$$

Weather in Albany

In this case, *Weather in LA* we call Event A and B are independent.

□ Consider again Rolling Two 6-sided Dice and Recording their Sum.

Sum	2	3	4	5	6	7	8	9	10	11	12
Probability	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

$$P(\underbrace{\text{Second Die is Even}}_A \mid \underbrace{\text{Die sum is 4}}_B)$$

$$= \frac{P(A \cap B)}{P(B)}$$

$$= \frac{\frac{1}{36}}{\frac{1}{12}} = \frac{1}{3}$$

$$P(B) = \frac{1}{12}$$

$$P(A \cap B) = P(\{2, 2\}) = \frac{1}{36}$$

$$P(\underbrace{\text{Sum is } \geq 8}_A \mid \underbrace{\text{One Die is } \geq 4}_B)$$

$$= \frac{P(A \cap B)}{P(B)}$$

$$P(B) = \frac{\quad}{36}$$

1	1
1	2
1	3
1	4
1	5
1	6
2	1
2	2
	...